

# Dr. Satveer Singh

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📄 ResearchGate

🎓 Google Scholar

🌐 LinkedIn



## Employment History

- Oct 2025 – till now **Assistant Professor**  
Institute of Engineering and Technology  
**JK Lakshmipat University, Jaipur**
- Feb 2024 – Sept 2025 **Assistant Professor**  
Dept. of IoT and Intelligent Systems  
**Manipal University Jaipur**
- Feb 2019 – July 2019 **Junior Research Fellow**  
Scientific Analysis Group  
**Defence Research and Development Organization (DRDO), Delhi**
- Sept 2016 – Jan 2019 **Assistant Professor**  
Dept. of Computer Science and Engineering  
**Maharani Girls Engineering College, Jaipur**

## Education

- Aug 2019 – July 2024 **Ph.D. (Computer Science and Technology)**  
**Jawaharlal Nehru University, New Delhi, India**  
School of Computer and Systems Sciences  
Thesis title: *Fog Device Deployment and IoT Workload Distribution in Fog-integrated Cloud: A Metaheuristic Approach*  
Course Work CGPA: 8.0
- July 2014 – Aug 2016 **M.Tech. (Computer Science and Engineering)**  
**Sant Longowal Institute of Engineering and Technology, Punjab, India**  
Dept. of Computer Science and Engineering  
Thesis title: *Strategies for Optimum Resource Allocation in Cloud*  
CGPA: 8.69
- July 2009 – July 2013 **B.Tech. (Computer Science and Engineering)**  
**Rajasthan Technical University, Rajasthan, India**  
Aravali Institute of Technical Studies, Udaipur, Rajasthan  
Major Project: *The Secured Ignition System*  
Percentage: 64.70

## Courses Taught

- |                                       |                                              |
|---------------------------------------|----------------------------------------------|
| C Programming                         | Procedural programming, memory management.   |
| Python Programming                    | Object-oriented programming, data scripting. |
| Data Structures                       | Abstract types, standard implementations.    |
| DAA (Design & Analysis of Algorithms) | Algorithm paradigms, complexity analysis.    |
| Computer Networks                     | Network protocols, socket programming.       |
| Edge, Fog & Cloud Computing           | Virtualization and resource scheduling.      |
| Machine Learning                      | Supervised, unsupervised, neural networks.   |

## Research Publications

### Journal Articles

- 1 R. Keshri, **S. Singh**, and D. P. Vidyarthi, "Fog Device Deployment for Enhanced Edge Communication and Network Cohesion using JAYA," *Journal of Grid Computing*, vol. 24, no. 2, p. 12, 2026.  DOI: 10.1007/s10723-026-09832-x.
- 2 **S. Singh**, H. Dureja, and R. Singh, "A multilingual smart farming system for real time agricultural decisions using machine learning," *Discover Applied Sciences*, vol. 8, no. 6, p. 613, 2026.  DOI: 10.1007/s42452-026-08597-y.
- 3 **S. Singh**, E. E. Sham, and D. P. Vidyarthi, "Optimizing Fog Device Deployment for Maximal Network Connectivity and Edge Coverage using Metaheuristic Algorithm," *Future Generation Computer Systems (FGCS)*, vol. 157, pp. 529–542, 2024.  DOI: 10.1016/j.future.2024.04.010.
- 4 **S. Singh**, E. E. Sham, and D. P. Vidyarthi, "Optimizing Workload Distribution in Fog-Cloud Ecosystem: A JAYA based Meta-heuristic for Energy-Efficient Applications," *Applied Soft Computing Journal*, vol. 154, pp. 1–18, 2024.  DOI: 10.1016/j.asoc.2024.111391.
- 5 **S. Singh** and D. P. Vidyarthi, "A Hybrid Model using JAYA-GA Metaheuristics for Placement of Fog Nodes in Fog-integrated Cloud," *Journal of Ambient Intelligence and Humanized Computing (JAIHC)*, pp. 3035–3052, 2024.  DOI: 10.1007/s12652-024-04796-w.
- 6 **S. Singh** and D. P. Vidyarthi, "FNSS: A Heuristics for Fog Node Site Selection," *SN Computer Science*, vol. 5, no. 146, pp. 1–18, 2024.  DOI: 10.1007/s42979-023-02468-8.
- 7 **S. Singh** and D. P. Vidyarthi, "PMRNA: Parameter Matching of Realtime and Non-Realtime Applications for Resource Provisioning in Fog-integrated Cloud," *Concurrency and Computation: Practice and Experience*, pp. 1–19, 2024.  DOI: 10.1002/cpe.8145.
- 8 **S. Singh** and D. P. Vidyarthi, "Fog Node Placement using Multi-Objective Genetic Algorithm," *International Journal of Information Technology (IJIT)*, vol. 16, no. 2, pp. 713–719, 2023.  DOI: 10.1007/s41870-023-01530-1.
- 9 **S. Singh** and J. Singh, "A Survey on Resource Allocation Strategies in Cloud," *International Journal of Computer Science and Engineering Technology (IJCSET)*, vol. 7, pp. 185–190, 2016.  URL: <https://www.ijcset.com/docs/IJCSET16-07-05-029.pdf>.

### Conference Proceedings

- 1 R. Keshri, **S. Singh**, and D. P. Vidyarthi, "Communication-Aware Fog Device Deployment in Edge-Fog-Cloud Systems Using Jaya Algorithm," in *International Conference on Data Science and Artificial Intelligence*, Springer, 2025, pp. 73–82.  DOI: 10.1007/978-981-96-9083-1\_6.
- 2 T. Vats and **S. Singh**, "Ev charging slot scheduling: Integrating advanced scheduling algorithms with sdn—a case study," in *International Conference on Next-Generation Networks and Deployable Artificial Intelligence*, Springer, 2025, pp. 116–126.  DOI: 10.1007/978-3-032-15398-2\_9.
- 3 **S. Singh** and D. P. Vidyarthi, "Designing Fog Device Network for Digitization of University Campus," in *International Conference on Soft Computing and its Engineering Applications*, **Springer, SCOPUS**, 2022, pp. 123–134.  DOI: 10.1007/978-3-031-27609-5\_10.
- 4 **S. Singh** and D. P. Vidyarthi, "Task Scheduling in Fog-integrated Cloud Environment," in *7th International Istanbul Scientific Research Congress*, Istanbul, 2021, pp. 701–709, ISBN: 978-605-71167-1-0.
- 5 **S. Singh** and B. Singh, "Optimum Resource Allocation Approach in Cloud," in *International Conference on Advanced Communication Control and Computing Technologies (ICACCCT)*, **IEEE, SCOPUS**, 2016, pp. 600–605.  DOI: 10.1109/ICACCCT.2016.7831711.

## Skills

|                        |                                                                                                     |
|------------------------|-----------------------------------------------------------------------------------------------------|
| Area of Specialization | Cloud Computing, Fog Computing, Internet of Things, and Machine Learning.                           |
| Languages              | Strong Reading, Writing and Speaking Competencies in English and Hindi.                             |
| Coding                 | C · Python · OPL (Optimization Programming Language)                                                |
| Simulators             | CloudSim · iFogSim · CPLEX STUDIO IDE 20.1.0 (IBM)                                                  |
| Misc.                  | Academic Research · Teaching · Training · Consultation · $\text{\LaTeX}$ Typesetting and Publishing |

## Achievements

|                               |                                                          |
|-------------------------------|----------------------------------------------------------|
| Aug 2021 – Feb 2024           | UGC-SRF (Senior Research Fellowship)                     |
| Aug 2019 – Aug 2021           | UGC-JRF (Junior Research Fellowship)                     |
| Feb 2019 – July 2019          | DRDO-JRF (Junior Research Fellowship)                    |
| July 2014 to July 2016        | GATE Scholarship (Graduate Aptitude Test in Engineering) |
| 2014 to 2020                  | GATE Qualified Seven Times                               |
| Nov 2017, June 2020           | UGC-NET/JRF Qualified Two Times                          |
| July 2016, Jan 2017, Dec 2018 | UGC-NET Qualified Three Times                            |

## Workshop/Conference/Summer School/FDP

|      |                                                                                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | <b>Two Days Faculty Development Program (FDP) on Rethinking Learning Outcomes and Assessments for Programming Courses in the GenAI Era</b> , April 03-04, Organized by <i>Institute of Engineering &amp; Technology at JK Lakshmipat University, Jaipur</i> .                           |
|      | <b>Session Chair</b> at the <b>IEEE International Conference on AI Engineering and Innovation (AIEI 2026)</b> , March 26-28, Organized by <i>National Institute of Technology, Jamshedpur, India</i> and <i>University of South Dakota, USA</i> .                                       |
| 2025 | <b>Five Days Faculty Development Program on Mastering Cloud Computing Essentials with AWS</b> , March 03-07, Hosted by <i>NITTTR, Chandigarh</i> .                                                                                                                                      |
| 2024 | <b>Five Days Faculty Development Program on Developing the Future Workforce: Designing an AI and Analytics curricula</b> , April 15-19, Hosted by <i>SAS</i> .                                                                                                                          |
|      | <b>Three Days Faculty Development Program by Dale Carnegie on Train The Trainer (TTT)</b> , Feb 19-21, Hosted by <i>Manipal University, Jaipur, India</i> .                                                                                                                             |
| 2023 | <b>Training Workshop on IoT Organized by 5G Indigenous Test Bed O and M Team</b> , 31st Oct, Hosted by <i>Indian Institute of Technology, Delhi, India</i> .                                                                                                                            |
|      | <b>One Week Short Term Course on Machine Learning and its Applications in IoT, Computer Vision and Cloud Computing (MICC-2023)</b> , July 10-14, Hosted by <i>National Institute of Technology, Jamshedpur, India</i> .                                                                 |
|      | Presented Paper <b>Fog Node Placement using Multi-Objective Genetic Algorithm</b> in <b>International Conference on Emerging Computational Intelligence (ICECI-2023)</b> , Feb 11-12, Hosted by <i>Aligarh Muslim University, Aligarh, India</i> .                                      |
| 2022 | <b>One-Week Workshop on Blockchain Technology: Architecture and Applications</b> , Nov 28- Dec 04, Hosted by <i>Indian Institute of Information Technology, Kota, India</i> .                                                                                                           |
|      | Presented Paper <b>Designing Fog Device Network for Digitization of University Campus</b> in <b>International Conference on Soft Computing and its Engineering Applications</b> , Dec 09-10, Hosted by <i>Charotar University of Science and Technology (CHARUSAT), Changa, India</i> . |
| 2021 | <b>Online Workshop on Cyber Security and Digital Forensics</b> , Aug 26- 30, Hosted by <i>Aligarh Muslim University, Aligarh, India</i> .                                                                                                                                               |

## Workshop/Conference/Summer School/FDP (continued)

-  Presented Paper **Task Scheduling in Fog-Integrated Cloud Environment** in **7th International Istanbul Scientific Research Congress**, Dec 18-19, Hosted by *International Science Culture and Youth Work Association, Turkey*.
- 2016   Presented Paper **Optimum Resource Allocation Approach in Cloud** in **International Conference on Advanced Communication Control and Computing Technologies (ICAC-CCT)**, May 25-27, Hosted by *Syed Ammal Engineering College, Tamilnadu, India*.
- 2012   **Four-week Summer Training on “Basic Telecom”** at *Bharat Sanchar Nigam Limited, Udaipur, India*.
- 2011   Participated in **National Workshop on Networking and Cyber Security**, Oct, Hosted by *Aravali Institute of Technical Studies, Udaipur, India*
-  Participated in **Workshop of 3D Animation and Multimedia**, May, Hosted by *Aravali Institute of Technical Studies, Udaipur, India*

## References

### Prof. Deo Prakash Vidyarthi

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Jawaharlal Nehru University  
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### Prof. Birmohan Singh

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Dept. of Computer Science and Engineering  
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